

AI-POWERED TRAVEL PLANNER

Multi-Step Prompt Workflow for Personalized Travel Planning

Mini Assessment – Prompt Workflow Design

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1. Project Objective

The objective of AI-Powered Travel Planner is to design a structured, multi-step prompt workflow that takes a traveler’s preferences and trip information as input, understands the travel requirements, analyzes destinations and activities, evaluates practical constraints, identifies gaps or risks, improves the itinerary, and produces a concise personalized travel preparation report. The workflow is completed through prompts in an AI interface without building application code.

2. Problem Statement

Travelers often find it difficult to create a complete itinerary that properly matches their destination, budget, available time, interests, transportation needs, accommodation preferences, and travel dates. They may overlook important attractions, travel time, costs, booking considerations, or practical constraints. A single AI prompt may provide general suggestions, but it does not always provide a systematic process for understanding the trip, analyzing requirements, evaluating options, identifying gaps, optimizing the itinerary, and preparing for the journey.

3. Workflow Architecture

INPUT → STEP 1 → STEP 2 → STEP 3 → STEP 4 → STEP 5 → STEP 6

Travel Preferences + Destination Information → Travel Requirement Understanding → Destination & Activity Analysis → Itinerary Match Evaluation → Budget & Practicality Assessment → Itinerary Optimization → Final Travel Report

4. Workflow Steps and Expected Outputs

STEP	PURPOSE	EXPECTED OUTPUT
1. Travel Requirement Understanding	Extract destination, dates, number of travelers, budget, interests, transport preferences, accommodation preferences and special requirements.	Structured traveler profile with verified trip information.
2. Destination & Activity Analysis	Break down the destination into suitable attractions, activities, neighborhoods, travel options and relevant experiences.	Structured destination and activity profile.
3. Itinerary Match Evaluation	Compare the traveler profile with destination options and identify strong and weak matches.	Match assessment, suitable activities, conflicts and evidence mapping.
4. Budget & Practicality	Identify estimated cost areas,	Budget/practicality report, risk

Assessment	travel-time issues, scheduling conflicts, availability considerations and practical constraints.	points and improvement areas.
5. Itinerary Optimization	Improve the itinerary while preserving the traveler's actual preferences, dates and constraints.	Optimized daily itinerary, activity ordering and planning suggestions.
6. Final Travel Preparation Report	Combine validated findings into a concise travel-ready report.	Final itinerary, priority recommendations, budget overview, preparation checklist and contingency plan.

5. Detailed Prompt Design

STEP 1 — Travel Requirement Understanding

You are a Travel Information Analyzer.

Analyze the provided travel request and create a structured traveler profile.

Identify:

1. Destination and preferred locations
2. Travel dates and trip duration
3. Number of travelers
4. Budget range
5. Interests and preferred activities
6. Transportation preferences
7. Accommodation preferences
8. Food or cultural preferences
9. Special requirements or restrictions, if mentioned

Travel Request:

[PASTE TRAVEL REQUEST HERE]

STEP 2 — Destination & Activity Analysis

You are a Destination and Activity Analyzer.

Analyze the provided destination information and identify:

10. Destination highlights
11. Recommended attractions and activities
12. Suitable neighborhoods or areas
13. Transportation options
14. Possible day-trip or nearby options
15. Activities matching the traveler's interests
16. Important planning considerations
17. Relevant timing or scheduling considerations

Do not add requirements or preferences that are not present in the traveler profile. Return the analysis using clear headings and concise bullet points.

Destination Information:

[PASTE DESTINATION INFORMATION HERE]

STEP 3 — Travel Match Evaluation

You are a Travel Match Evaluator.

Using the traveler profile from Step 1 and the destination analysis from Step 2, evaluate how well the available options match the trip requirements.

Provide:

18. Overall match score from 1–10
19. Strongly matching activities and locations
20. Matching experiences and transportation options
21. Unsuitable, missing or weak options
22. Traveler-requirement-to-itinerary evidence mapping
23. Three highest-priority improvements

Base the evaluation only on the supplied information. Do not invent preferences, bookings or availability.

STEP 1 OUTPUT:

[PASTE STEP 1 OUTPUT HERE]

STEP 2 OUTPUT:

[PASTE STEP 2 OUTPUT HERE]

STEP 4 — Budget & Practicality Assessment

You are a Travel Budget and Practicality Assessment Assistant.

Using the travel-match evaluation, identify the most important budget and planning gaps.

Evaluate:

24. Accommodation and transportation cost areas
25. Activity and entry-fee considerations
26. Daily spending considerations
27. Travel-time and distance issues
28. Scheduling conflicts or overly packed days
29. Potential availability or booking considerations
30. Three practical actions to improve the itinerary

Do not present uncertain prices or availability as confirmed facts. Clearly identify estimates and items that require verification.

STEP 3 OUTPUT:

[PASTE COMPLETE STEP 3 OUTPUT HERE]

STEP 5 — Itinerary Optimization

You are an Itinerary Optimization Assistant.

Improve the traveler’s itinerary based on the previous analysis.

Preserve the traveler’s actual destination, dates, preferences and constraints. Do not invent bookings, prices, reservations or experiences.

Provide:

- 31. Improved day-by-day itinerary
- 32. Better activity ordering based on location and time
- 33. Balanced activity and rest periods
- 34. Transportation suggestions
- 35. Budget-control suggestions
- 36. Alternative options for important activities
- 37. Formatting/content improvement suggestions
- 38. Explanation of how the changes address the identified gaps

PREVIOUS ANALYSIS:

[PASTE STEP 1, STEP 2, STEP 3 AND STEP 4 OUTPUTS HERE]

STEP 6 — Final Travel Preparation Report

You are a Final Travel Preparation Report Generator.

Using all previous analyses, generate a concise travel-ready report.

Include:

- 39. Destination and travel dates
- 40. Overall itinerary match score
- 41. Top recommended activities and locations
- 42. Main planning or budget gaps
- 43. Highest-priority itinerary improvements
- 44. Recommended local experiences or learning topics
- 45. Transportation and accommodation preparation topics
- 46. Optimized day-by-day itinerary
- 47. Three immediate action items
- 48. Travel checklist and simple contingency plan

Do not invent facts, bookings, prices, availability or statistics. Clearly distinguish between confirmed information, estimates and recommendations.

PREVIOUS WORKFLOW OUTPUTS:

[PASTE ALL PREVIOUS STEP OUTPUTS HERE]

6. Prompt Chaining & Information Flow

FROM → TO	PURPOSE
Step 1 → Step 2	Traveler profile supports destination and activity analysis
Step 2 → Step 3	Destination options support travel-match evaluation
Step 3 → Step 4	Match findings reveal budget and practicality gaps
Step 4 → Step 5	Gaps guide itinerary improvements

7. Demonstration Example

INPUT EXAMPLE

Travel Request: A traveler wants a 5-day city trip with a moderate budget, prefers cultural attractions, local food, scenic places and convenient public transportation.

Destination Information: The destination offers museums, historical landmarks, local markets, scenic viewpoints, public transport and several neighborhoods with different experiences.

Expected Result

A structured travel match score, suitable activities, planning gaps, practical budget recommendations, an optimized day-by-day itinerary and a focused travel preparation checklist.

8. Expected Learning Outcome

This workflow demonstrates how prompt engineering can decompose a complex travel-planning task into manageable AI interactions. It highlights sequential prompting, contextual reuse, structured evaluation, gap analysis, iterative improvement and final content generation without requiring software development or code.

9. Conclusion

AI-Powered Travel Planner transforms travel preferences and destination information into a structured, evaluated and travel-ready preparation plan through six connected prompt stages. By separating requirement understanding, destination analysis, matching, practicality assessment, itinerary optimization and final reporting, the workflow provides a clear, professional and reproducible approach to AI-assisted travel planning.